A Preliminary Study of Users' Experiences of Meditation in Virtual Reality

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Abstract

This poster describes a between-groups study (n=24) exploring the use of virtual reality (VR) for facilitating focused meditation. Half of the participants were exposed to a meditation session combing the sound of a guiding voice and a visual environment including virtual objects for the participants to focus on. The other half of the participants were only exposed to the auditory guide. The participants' experience of the sessions was assessed using self-reported measures of perceived concentration, temporal duration, stress reduction, and comfort. Interestingly, no statistically significant differences were found between the two conditions. This finding may be revealing in regards to the usefulness of VR-based meditation.

Index Terms: H.1.2 [Information Systems]: User/Machine Systems—Human factors; I.3.7 [Computer Graphics]: Three-Dimensional Graphics and Realism—Virtual Reality;

1 Introduction

During the second half of the twentieth century, the field of psychology was heavily focused on the study and treatment of mental illnesses and disorders [1]. However, a relatively recent strand of psychology, dubbed *Positive Psychology*, has sought to return focus to the optimization of human experience (e.g., increasing happiness, success and virtue) [3]. Positive psychology interventions are a collection of approahes intended to increase individuals' well-being by cultivating positive feelings, behaviours and cognition [5]. These methods include, but are not limited to, practising optimistic thinking, replaying positive experiences, and meditation. The current poster focuses on the use of Virtual Reality (VR) for facilitating meditation. Meditation has generally been defined as a "family of techniques which have in common a conscious attempt to focus attention in a non-analytical way, and an attempt not to dwell on discursive, ruminating thought." [4]

A large body of work has explored the use of VR for treatment of mental illness and disorders; e.g., acrophobia, aviophobia, arachnophobia, body image disturbances, and post-traumatic stress (see [2] and references therein). Previous research has explored technology-assisted meditation (e.g., [6]), and there exist commercial VR applications explicitly designed for meditation as well as applications that provide meditation-like experiences; e.g, Guided Meditation VR (www.guidedmeditationvr.com) and Perfect Beach (www.ndreams.com). However, to the best of the authors' knowledge, little or no research has addressed whether VR can be used to augment and improve existing meditation practices.

The study documented in the current poster constitutes an initial step towards addressing this question. Specifically, the aim of the study was to explore how guided meditation within a virtual environment (VE) would compare to a regular meditation session in terms of the users' perception of their ability to concentrate, the experienced temporal duration of the session, the perceived reduction in stress, and the experience of comfort during the session.

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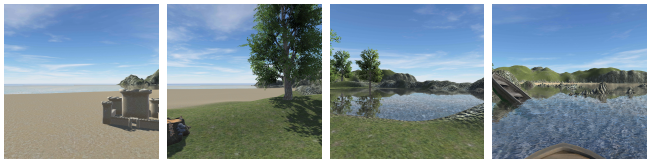

Figure 1: The four VEs made available to the participants. From the left: beach, grass, lake, and boat.

2 Method and Materials

A total of 24 participants (18 males and 6 females), between the ages of 18-33 (M=22.1, SD=3.3) took part in the study. All participants comprised university students from Aalborg University Copenhagen. The study relied on a between-groups design and compared two conditions:

- *Audio*: The participants were asked to complete a 10 minute guided meditation session by following a series of recorded instructions.
- *VR*: The participants completed a guided meditation session while exposed to one of four scenic VEs and listening to the same audio guide used for the Audio condition.

The participants were assigned to the two conditions in a pseudo-randomized manner; i.e., the individual participant would be randomly assigned to one of the two conditions unless it entailed that the gender distribution across conditions was not maintained. The participants assigned to the VR condition were free to choose one of four virtual environments: (a) *Beach*, (b) *Grass*, (c) *Lake*, and (d) *Boat* (Figure 1). The design of the four VEs was informed by a small user study. Thirty undergraduate students from Aalborg University Copenhagen were shown 15 pictures of different scenic environments and were asked to rate them based on how they relaxing they appeared. The central attributes of the four environments (e.g., the presence of water, mountains, and vegetation) were based on the pictures that scored the highest. These attributes might not be equally appealing to members of another demographic. However, since the final evaluation were to be performed on a similar user group this was not considered to be an issue. In order to give the participants objects of interest to focus on a series of *focus objects* were placed close to the center of the initial viewing direction (e.g., a capsized rowing boat and a sand castle). Specifically, the objects were intended to support visually focused meditation where attention is directed at an external object or event, rather than internal sensations. If the participants' gaze started to wander (crudely approximated from their head orientation) then a subtle bell sound would be played. The bell sound was intended to help draw the participants' attention back to the current object of focus. The soundscapes of the VEs comprised soothing, natural sounds congruent

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Table 1: mean ranks and medians pertaining to the four self-reported measures used for the study

	Mean ranks		Medians	
	Audio	VR	Audio	VR
(a) Perceived concentration	13.79	11.21	5.5	5.5
(b) Perceived duration	13.38	11.62	4.5	4.5
(c) Perceived stress reduction	10.33	13.67	4.0	5.0
(d) Perceived comfort	12.08	12.92	5.5	6.0

with the individual environments. The audio guide was designed with the aid of an expert in psychotherapy and inspired by the audio guide included in the commercial meditation application Headspace (www.headspace.com).

The VEs were developed in Unity 3D and delivered using a Samsung Gear VR. The auditory feedback was played back using a pair of circumarual headphones. The participants were seated throughout the meditation session and the only instructions given were that they should attempt to follow the audio guide.

The study solely relied on subjective measures of the participants experience of the VR and Audio meditation sessions. After completing the meditation session the participants filled out a questionnaire including 7-point Likert-type items (1-7) asking them to evaluate their experience in regards to the following four dimensions: (a) *perceived concentration* (to what degree where they able to maintain concentration during the session), (b) *perceived duration* (to what degree was the session perceived as too long), (c)*perceived stress reduction* (to what degree did the session help them feel less stressed), and (d) *perceived comfort* (to what degree did the feel comfortable during the session).

3 Results

The data obtained from the four questionnaire items was treated as ordinal, and Mann-Whitney U tests were used to identify differences in scores between the VR and the Audio group. In no cases were the distributions of scores for the two conditions similar, as assessed by visual inspection. Thus, the comparisons were focused on mean ranks rather than medians. Figure 2 illustrates the distribution of ratings across the four questionnaire items, and Table 1 summarizes the corresponding mean ranks and medians. The pairwise comparisons using Mann-Whitney U tests revealed no statistically significant differences between conditions: perceived concentration ($U = 87.5, z = .355, p = .378$), perceived duration ($U = 82.5, z = .502, p = .551$), perceived stress reduction ($U = 46.6, z = .123, p = .143$), and perceived comfort ($U = 67.0, z = .756, p = .799$). Out of the four VEs made available to the 12 participants in the VR group, five chose the boat, four chose the beach, three chose the lake, and none chose grass.

4 Conclusion

The four self-reported measures did as suggested not reveal any statistically significant differences between the scores accompanying exposure to the two conditions. However, it is worth noting that the scores related to perceived stress reduction yielded a slight, albeit insignificant ($p = .143$), indication that the VR may have positively influenced their experience of the meditation session; i.e., this measure yielded the largest difference between mean ranks and the lowest p-value. Nevertheless, even though null results do not justify the conclusion that no differences exist, it does seem likely that the VE did not have any noteworthy effect on the participants' experience of the meditation session. A plausible explanation is that, as long as the setting is sufficiently calm, which it arguably was in both conditions, then novice meditators might be able to

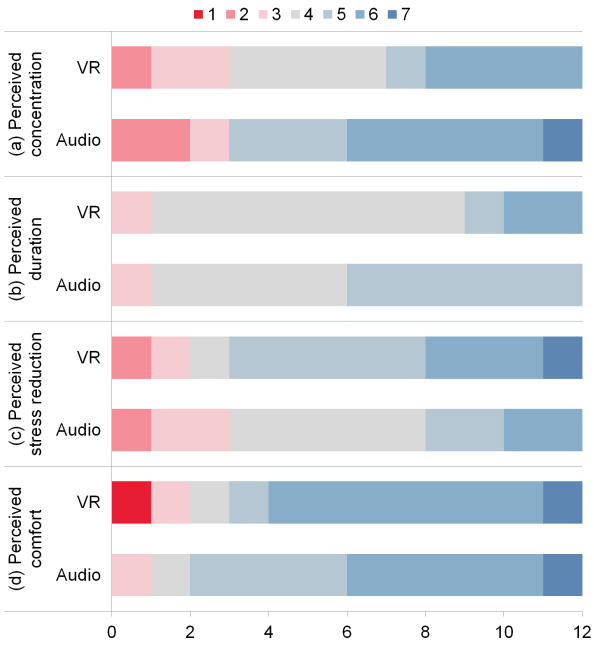

Figure 2: Visualization of the distribution of ratings (1-7) across the four self-reported measures used for the study.

take steps towards focusing their attention in a non-analytical way, and avoid dwelling on discursive, ruminating thought. It should be stressed that the limited sample size of the study and the particular design and implementation also may have decreased the likelihood of identifying potential benefits of VR-based meditation. For example, the focus objects and bell sound intended to support visually focused meditation could have had an adverse effect; i.e., unnecessarily constraining and distracting the user. Moreover, the limited sample size prevents us from concluding whether the different VEs were perceived as equally beneficial. Thus, we cannot rule out that certain types of VR-based meditation will be perceived more positively than a purely auditory guide. Moreover, it should be highlighted that the current study does not reveal anything about whether VR-based meditation indeed influences concentration or has additional benefits with respect to increasing the well-being of the individual user. Longitudinal studies and studies relying on alternative metrics (e.g., physiological measures) are needed in order to explore such claims. Finally, it seems possible that VR-based meditation may be accompanied by indirect benefits to well-being; e.g., it might make individuals more motivated to meditate and help sustain the practice over extended periods of time.

References

[1] K. Hefferon and I. Boniwell. *Positive psychology: Theory, research and applications*. McGraw-Hill Education (UK), 2011.

[2] G. Riva. Virtual reality in psychotherapy: review. *Cyberpsychology & behavior*, 8(3):220–230, 2005.

[3] S. Schueller. Positive psychology. *Encyclopedia of Human Behavior*, pages 140–147, 2012.

[4] D. H. Shapiro Jr. *Meditation: Self-regulation strategy and altered state of consciousness*. Transaction Publishers, 2008.

[5] N. L. Sin and S. Lyubomirsky. Enhancing well-being and alleviating depressive symptoms with positive psychology interventions: A practice-friendly meta-analysis. *Journal of clinical psychology*, 65(5):467–487, 2009.

[6] J. Vidyarthi and B. E. Riecke. Interactively mediating experiences of mindfulness meditation. *International Journal of Human-Computer Studies*, 72(8):674–688, 2014.

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